Network Port Scanner

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PROBLEM STATEMENT

Existing port scanning tools are often complex, command-line based, or require advanced technical knowledge. There is a need for a simple, lightweight, and user-friendly GUI-based TCP port scanner that allows users to efficiently scan ports, view results in real time, identify services, and save outputs without requiring complex setup or external dependencies.

Project Description

Network Port Scanner GUI is a lightweight application developed using Python and Tkinter that enables users to scan TCP ports on a target system through a simple graphical interface. The project is designed to make network scanning accessible to beginners while still providing efficient performance.

The application allows users to enter a target IP address or hostname along with a range of ports to scan. It uses multi-threading to perform fast and efficient scanning, displaying open ports in real time. The tool also identifies common services running on well-known ports, helping users understand the network status more easily.

Additionally, the system includes features such as a progress bar, elapsed time tracking, the ability to stop scans midway, and an option to save results for future reference. Being cross-platform and requiring no external libraries, the project is simple to deploy and use.

Overall, this project serves as a practical tool for network analysis, learning, and authorized security testing, combining usability with performance in a clean GUI environment.

WHO ARE THE END USERS?

- Students and beginners in networking/cybersecurity
- Cybersecurity enthusiasts and ethical hackers
- System administrators
- Network administrators
- IT professionals and testers
- Educational institutions (for lab use)

Technology Used

- Python – Core programming language
- Tkinter – GUI development
- Socket Programming – For TCP port scanning
- Multithreading (threading module) – To improve scanning speed
- Time module – For tracking elapsed time
- File Handling – To save scan results in .txt format

RESULTS

Port Scanning Report

1. Target Information

- **Target IP Address:** 192.168.1.1
- **Scan Range:** Ports 1 – 1024
- **Scan Type:** Basic TCP Port Scan (Well-known ports)

2. Scan Summary

The scan identified several open ports on the target system. Open ports indicate active services that may be accessible over the network.

3. Open Ports and Services

Port Number	Service	Status	Description
21	FTP	Open	File Transfer Protocol used for file sharing
53	DNS	Open	Domain Name System for resolving domain names
80	HTTP	Open	Web server (unencrypted communication)
443	HTTPS	Open	Secure web server (encrypted communication)

4. Service Analysis

Port 21 – FTP

- Used for file transfer between client and server
- May allow anonymous login if not secured
- **Risk:** Weak authentication or misconfiguration can lead to unauthorized access

Port 53 – DNS

- Handles domain name resolution
- Typically used by routers or DNS servers
- **Risk:** DNS amplification attacks or information leakage

Port 80 – HTTP

- Standard web service port
- Transmits data in plain text
- **Risk:** Vulnerable to sniffing, man-in-the-middle attacks

Demo Link

RESULTS

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Port 443 – HTTPS

- Secure web communication using SSL/TLS
- Encrypts data between client and server
- **Risk:** Misconfigured SSL certificates or outdated protocols

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5. Observations

- The target system appears to be a **network gateway or router**, as indicated by DNS and web services.
- Both **HTTP (80)** and **HTTPS (443)** are active, suggesting a web-based management interface.
- Presence of FTP (21) may indicate file-sharing capability or legacy configuration.

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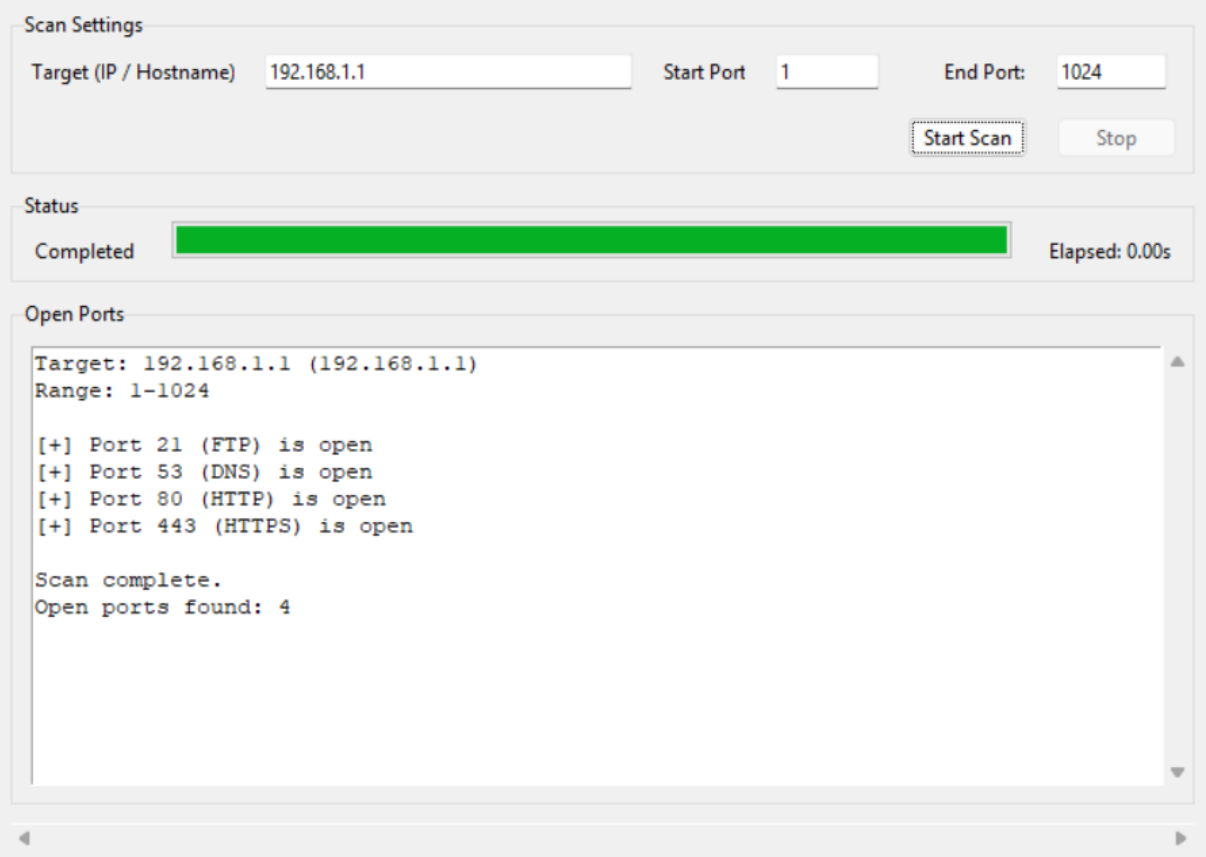
6. Security Recommendations

1. **Disable unused services**
   - If FTP is not required, disable port 21



Demo Link

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GitHub link

https://github.com/gokulraj-design/portscannervoisinternship.git

Thank you

By

GOKULRAJ KR